

Figure 5 — Standalone Interactive HTML Supplementary Material (~400 KB, offline-capable)

A. Biogeochemical Cycle Panel

- D3.js force simulation
- 4-element quadrants (As / N / S / Fe)
- Draggable nodes; hover tooltips
- Chemistry coupling anchors
- Click → gene-level detail
- SVG export

B. Hypothesis Scoring Panel

- 6-claim YAML scoring table
- Sortable by score / null_p / label
- Per-claim diagnostic 5-tuple
- null_p distribution histogram
- Saltelli $\pm 20\%$ weight bands
- Per-group toggle (CK / A / B)

C. Cross-Group Comparison Panel

- Pathway × Group long-format heatmap
- Top-MAG keystone tracking
- Group color: standardized RGB
- Sort: contribution / top-MAG switch
- Element filter (As / N / S / Fe)
- Element / keystone change

D. Parameter Audit Panel

- All S1 sensitivity rows
- All S2 (pathway × env) cells
- Permutation null distribution
- Saltelli weight-perturb data
- JSON download (machine-readable)
- Reproducibility certified